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A Cost Benefit Analysis Of Pulse Jet Actuators In The Slat Cut-Out Region

BEng PROJECT FINAL DISSERTATION 2022-23

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Summary

As engines become greater in size due to higher fuel efficiencies and reduced CO2 emissions, they take up more room near the wing's leading edge. As a result, the slat cut-out region increases. This reduces the effectiveness of slats overall, causing a reduction in lift and earlier onset of stall when the slats are extended. Pulse jet actuators are being experimented in the cut-out region to investigate how they can recover lift in this region. However, these devices come at an energy cost which needs to be evaluated against the aerodynamic performance benefits they create. The aim of this study is to find how cost-effective pulse jet actuators are in the slat cut region in terms of power consumption. To achieve this, a CAD model was created to find the dimensions of components inside the pulse jet system. Next, a MATLAB script was created to calculate pneumatic and electrical power consumptions to compare against the aerodynamic performance data that was collected in an external project, WingPulse (where pneumatic data was collected too). Meanwhile, a separate experiment testing a single solenoid from the pulse jet actuator system was conducted to obtain the electrical power consumption data. Then a cost benefit analysis of the pulse jet actuators was run to compare the aerodynamic performance increases directly against the power consumptions of the system. From this analysis, the most optimal tested configuration for the pulse jets was chosen. Additionally, analysis of the pressure head losses of the system was also conducted. It was discovered that the pulse jet

actuator system was most power efficient at high angles of attack and the lowest duty cycle of 20% which caused as much as a 24% increase in lift while having the smallest power consumption. Meanwhile, drag was also reduced the most at these high angles of attack but less efficiently, with the drag reduction reaching a maximum of 7% at the highest duty cycle. It was found that higher duty cycles caused the greatest net benefit in drag reduction when an equivalent drag, caused by the power consumption, was considered. Pressure head losses were found to increase at higher duty cycles, further reducing the efficiency of a higher duty cycle. The Y-junction and 90° turn in pipe caused the greatest head losses in the system.

Contents

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NOMENCLATURE

Symbol	Quantity	Unit
$C_{D_{Power}}$	Power consumption equivalent drag coefficient	-
$C_{D_{0\%}}$	Drag coefficient at 0% duty cycle	-
$C_{D_{x\%}}$	Drag coefficient at x% duty cycle	-
$C_{L_{0\%}}$	Lift coefficient at 0% duty cycle	-
$C_{L_{max}}$	Maximum lift coefficient	-
$C_{L_{x\%}}$	Lift coefficient at x% duty cycle	-
C_D	Drag coefficient	-
C_L	Lift coefficient	-
$D_{Exterior}$	Diameter of exterior pipe below the wing	m
D_P	Diameter of pipes inside the wing (pipe 1 and pipe 2)	m
D_{Power}	Power consumption equivalent drag	N
I_P	Peak current	A
L_1	Length of pipe 1	m
L_{2A}	Length of pipe 2A	m
L_{2B}	Length of pipe 2B	m
$P_{Electrical}$	Electrical power consumption	J/s
$P_{Pneumatic}$	Pneumatic power consumption	J/s
P_{total}	Total power consumption	J/s
Re_1	Reynolds number at pipe 1	-
Re_{2A}	Reynolds number at pipe 2A	-
Re_{2B}	Reynolds number at pipe 2B	-
S_{ref}	Reference surface area	m^2
V_∞	Freestream velocity	m/s
V_P	Peak voltage	V
V_{P1}	Velocity in pipe 1	m/s
V_{P2A}	Velocity at pipe 2A	m/s
V_{P2B}	Velocity at pipe 2B	m/s
V_T	Velocity at pressure tank inlet	m/s
V_{jet}	Jet velocity	m/s
c_μ	Momentum coefficient	-
ff_1	Friction factor of pipe 1	-
ff_{2A}	Friction factor of pipe 2A	-
ff_{2B}	Friction factor of pipe 2B	-
$k_{180°}$	Head loss coefficient of 180° turn in pipe	-
$k_{90°}$	Head loss coefficient of 90° turn in pipe	-
k_Y	Head loss coefficient of Y-junction in pipe	-
$\dot{m}$	Mass flow	kg/s

p_K	Pressure at the Kulite	N/m^2
p_T	Pressure at pressure tank inlet	N/m^2
q_∞	Dynamic pressure	kg/ms^2
ρ_∞	Freestream density	kg/m^3
ρ_K	Density at the Kulite	kg/m^3
ρ_T	Density at pressure tank inlet	kg/m^3
ΔC_D	Change in drag coefficient	-
Δp_1	Pressure head loss of pipe 1	N/m^2
$\Delta p_{180°}$	Pressure head loss of 180° turn in pipe	N/m^2
Δp_{2A}	Pressure head loss of pipe 2A	N/m^2
Δp_{2B}	Pressure head loss of pipe 2B	N/m^2
$\Delta p_{90°}$	Pressure head loss of 90° turn in pipe	N/m^2
Δp_Y	Pressure head loss of Y-junction in pipe	N/m^2
I	Momentum flux added by actuation system	N/m^2
R	Gas constant for dry air	J/kgK
S	Wing area	m^2
T	Temperature	K
VR	Velocity ratio	-
v	Kinematic viscosity	m^2/s

ACRONYMS

Acronym	**Meaning**
VR	Velocity Ratio
AoA	Angle of Attack

[redacted]

1. INTRODUCTION.

Commercial jet engines are continually designed with greater bypass ratios to improve the efficiency of the engine and lower fuel consumption [1]. With lower fuel consumptions, aircraft can meet the environmental goals, such as those of Flightpath 2050 that aim to reduce levels of CO2 and NOx emissions [2] – these are directly caused by the consumption of fuel in the jet engine. Reducing fuel consumption also reduces overall fuel costs, making air travel more profitable/cheaper for companies. Greater bypass ratios result in engines becoming bigger in size. As engine size increases, the engine needs to be mounted further up onto the wing to make sure there is ground clearance below the engine. This means moving the engine further into the wing and as a result the slat cut-out region becoming larger (as shown in Figure 1). This reduces the coverage of the slats along the wing, hence lowering the effectiveness of the slats overall. The pulse jet actuators are designed to replace the slats in this region to get back the lost effectiveness.

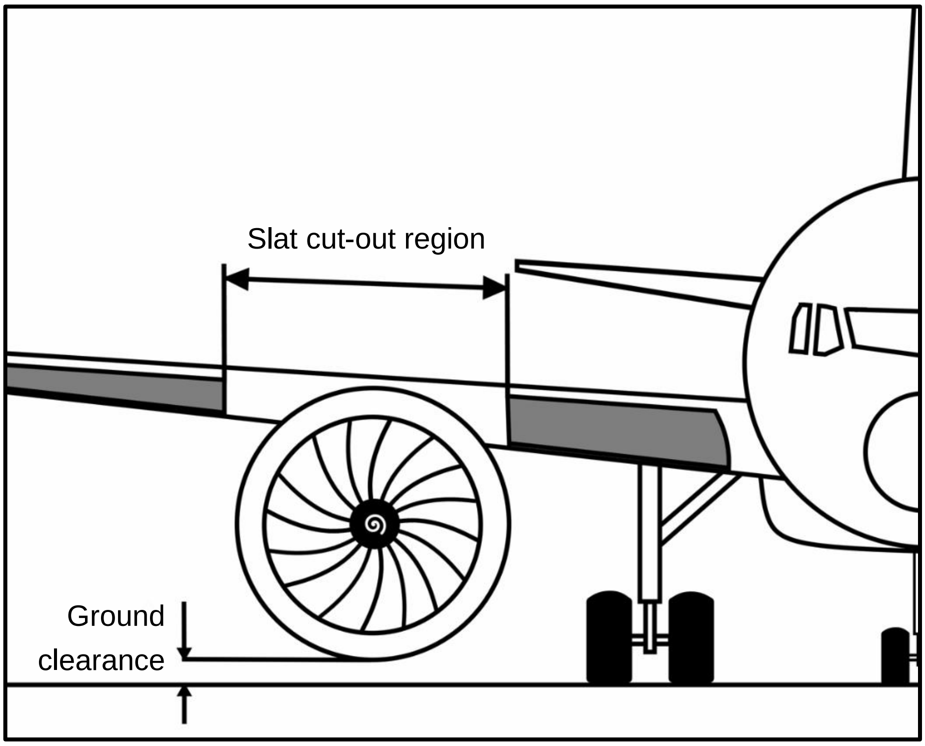

Figure 1 – Slat cut-region for an ultra-high bypass ratio engine.

Pulse jets delay the separation of the airflow along the boundary layer on the upper surface of the wing. Both the slats and pulse jets aim to delay the stall of the wing. The pulse jet actuator system comes at an energy and fuel consumption cost, which needs to be considered before making them a viable replacement/stand in for the slats in the cut-out region. The effectiveness of pulse jets in delaying the separation of the airflow will be tested in the WingPulse project [3] which this project is based on as to carry out the cost/benefit analysis.

The pulse jet actuator system will require both electric and pneumatic power to operate. In terms of active flow control, this is known as a hybrid system [4]. The aim of this project is to assess the power consumption of this pulse jet actuator system in the hybrid configuration by calculating the power draw from the pneumatic and electrical systems separately and comparing with the effectiveness of the system on flow control to find the most optimal and efficient pulse jet configuration. The cost/benefit analysis is focussed specifically on power consumption as it directly relates to the fuel consumption that is aimed to be reduced by the greater bypass ratio engines- power consumption of the pulse jet system is taken from the power the engine creates. The data from this project will be useful information for the real application of such a system in the future when the increasing size of the cut-out region becomes a problem to be concerned about.

1.1. Aim and Objectives

The aim of this project was to carry out a cost benefit analysis to study the energy consumption of using pulse jet actuators for active flow control in the slat cut-out region to facilitate Ultra High Bypass Ratio engines.

There are several project objectives:

- Create a drawing of the layout of the flow control system inside the wing and set up a CAD model of the system in the wing in Catia to obtain dimension measurements inside the flow control system.
- Develop a tool in MATLAB to calculate the energy and power consumptions at different configurations of duty cycle for the pulse jet actuators.
    - Procure and test the solenoid under the various conditions it would experience during the WingPulse project with varying duty cycles to obtain electrical power consumption data.
- Create a cost benefit analysis using the aerodynamic performance measurements from the WingPulse project, as well as the power consumption data that has been calculated.
- Identify the most optimal configuration for the slat cut-out region.

2. LITERATURE REVIEW.

Active flow control has been used in several applications for over half a century. Blowing is a type of active flow control that takes advantage of the Coanda effect- where air blown at a tangent to a curved surface would tend to follow along the surface- which moves the separation point further along the surface, delaying the stall [5].

The slat reduces the velocity of the boundary layer along the top surface of the aerofoil whilst increasing the velocity of the boundary layer along the bottom surface, thus reducing the pressure gradient between the two surfaces, and delaying the separation of the airflow [5]. This, together with what was mentioned on active flow control (in the form of blowing), suggests that an active flow control system has potential to be used as a replacement or as an addition to slats. However, according to another source, flow control at the leading edge (where the slats were typically located) required greater power than at the trailing edge as the local velocity in that region was greater [6]. This implied that active flow control in the slat cut-out region would have a higher power consumption as opposed to at the trailing edge, where the system has been historically placed.

It was declared that pulse jets could increase peak lift by 7% and maximum lift-to-drag ratio by 17% whilst decreasing mass flow by 80% (with a 20% duty cycle) over steady jet active flow control [7]. This suggested that, despite the higher power consumption of having the active flow system at the leading edge, the increased mass flow efficiency of the pulse jet could outweigh the increase in power demand in this region. An increased efficiency would be very important when comparing the power consumption against the increase in flight performance as power consumption needs to be minimised to be a viable, economical replacement in the slat cut-out region. Longitudinal vortices delay the separation of surface bounded flow because of the circulation of the vortices inside the boundary layer. These vortices energise the boundary layer by turbulent mixing which increases the boundary layer's momentum and flux. The increase in momentum increases the skin friction drag inside the boundary layer, which prevents the flow from slowing down and subsequently causing flow separation from the surface [7]. As the momentum is one of the main mechanisms behind the increased effectiveness of pulse jets, the momentum coefficient can be used to compare different configurations directly [8]:

$$c_\mu = \frac{I}{q_\infty * S_{ref}}$$ *Equation 1*

Where c_μ is the momentum coefficient, I is the momentum flux added by the actuation system, q_∞ is the dynamic pressure, and S_{ref} is the reference surface area. Furthermore, it was affirmed that pulsing significantly increases performance in flow control. Pulsing delays flow separation and the stall of the wing at higher angles of attack than when compared to the same mass flow of a steady jet. This means that the increased flight performance over power consumption is greater for pulsed jet flow compared to steady jet flow as the pulsed air required less mass flow per second than steady blown air so less pneumatic power was consumed from the reduced mass flow. The pulse jet also penetrated the boundary layer approximately 50% further than the steady jet flow and had a larger area of effect over the wing [9].

The performance of pulse jets heavily depends on the configuration of the variables of the pulse jet. One source conducted tests to find the most optimal setup, taking into consideration both the aerodynamic performance of the pulse jet and the energy/mass flow cost required. It was found that the optimum value for jet-to-freestream velocity ratios (VR) to be between 1.6-3.1, with efficiency of control decreasing above 2.5. These figures were similar to previous tests done on pulse jets [7]. As VR increased to 4.6, vortices had a tendency of lifting out of the boundary layer thus dissipating. However, this tendency could be reduced by increasing the duty cycle of the pulse from 25% to 50%. Simultaneously, increasing duty cycle while having a high VR would cause an increase in power consumption which needs to be minimised to create the biggest net benefit of using the pulse jets. At VRs

[redacted]

near 1, jet flow was non uniform from being disturbed from cross flow- increasing the VR reduced this cross flow and hence the jet flow became more uniform [7]. Velocity ratio is one of the most important parameters in active flow control [8] and is defined as:

$$VR = \frac{V_{jet}}{V_{\infty}} \qquad \text{Equation 2}$$

Where VR is velocity ratio, u_{jet} is the velocity of the jet flow, and u_{∞} is the freestream velocity. From all the different configurations tested, the most optimum configuration for decreasing the drag coefficient while increasing the lift coefficient at a minimum mass flow was a VR of 2, pulse frequency of 15Hz, and duty cycle of 20% [7]. When duty cycle is low however, the velocity of the jet decays at an exponential rate very close to the nozzle exit, which may be due to the high Reynolds number of the jet flow [10]. As duty cycle increases up to 80%, successive vortex rings move closer together and eventually interact with each other which results in the destruction of the vortex rings and subsequent turbulent flow is produced [10]. Frequency of the pulse is another important factor to consider in the pulse jet actuator [11]. F+ frequency is a non-dimensional frequency (also known as the reduced frequency) is commonly used in pulse jet flow which is defined by the pulse frequency divided by the natural frequency of the uncontrolled flow field [12]. By pulsing the jet at specific frequencies, the flow's inertia can be taken advantage of by turning off the pulsing until the airflow starts to separate from the boundary again. The specific frequency varies on the application so the correct excitation frequency would need to be determined in advance [11]. At 14 degrees of angle of attack, the drag coefficient was decreased by 10.6% and the maximum lift coefficient was increased by 6% over that of a steady jet flow control [7]. However, this configuration may not be consistently the most optimum in other experiments, as factors such as the aerofoil and angle of attack will affect the performance of the flow control. The data collected in this project will most likely vary from previous experiments [7], but this data remains useful as a rough estimate for performance expected from the pulse jets. At increasing duty cycles up to 60%, it was discovered that the strength of the vortices that the pulse jet created also increased [10]. This suggests that running a lower duty cycle might be more optimal when performance increase from the pulse jets is measured against their power consumptions in the cost benefit analysis.

3. Methodology

The methodology of this project consisted of two physical rigs and a computer program with an accompanying CAD model. The first rig was the WingPulse project rig which contained the wing with the pulse jet actuators in the slat cut-out region inside a wind tunnel. This was used to collect data on the lift and drag performance of the pulse jets and pressure and mass flow inside the pulse jet system. The second rig was the solenoid rig which was used to collect data on a single solenoid's voltage and current and thus power consumption, under the same conditions it would experience in the WingPulse project (data on the solenoid in WingPulse could not be obtained, as it lacked the sensors to collect solenoid performance data inside the wing). A CAD model of the WingPulse flow control system was created to aid the calculations in the cost benefit analysis, particularly those related to the dimensions of components inside the pulse jet system. A computer program was created which used data collected from the physical rigs to run the cost benefit analysis of the pulse jets.

3.1. Rig description

The WingPulse project was the main physical rig used and was utilised to collect the aerodynamic performance and pneumatic data. It consists of a wing containing a slat cut-out region and the flow control system inside a wing tunnel. The experiment was run at different configurations of duty cycle ranging from 20% to 70% in 10% increments while taking measurements at different angles of attack from 0° to 16° for all duty cycles. There were two cases of the experiment run:

Table 1 – The two cases of the experiment and their variables

Case	Pressure (bar)	Velocity Ratio	Pulse Frequency (Hz)	F+ frequency
Case 1	1.2	4.2	50	0.57
Case 2	1.9	5.0	44.1	0.50

The cost benefit analysis measured the performance against power consumption at 8° and 14° angles of attack. An additional 0% duty cycle configuration was run as a control variable set. Aerodynamic lift and drag data were collected to measure what effect the pulse jets had on the wing's lift and drag. A force balance was used to measure the lift enhancement and drag reduction at the different configurations of duty cycle and angle of attack. The freestream velocity V_∞ and density ρ_∞ in the wind tunnel were measured as 30m/s and 1.204kg/m³, respectively. The wing area was measured as 0.306m². The lift and drag coefficients were calculated from the force balance data.

The pipe system inside the WingPulse rig allowed the air to be pulsed through the wing from an external connection to the nozzles, which were located in the slat cut-out region. This caused the active flow control to occur on the wing. Pneumatic power was determined by the pressurized air that was put into the pipe system while electrical power controlled the duty cycle of the pulsed air using solenoids.

The pipe system consisted of the exterior pipe below the wing (at the root of the wing as the wing was mounted sideways) that connected the incoming pressurised air to the pressure tank. This pipe contained the Alicat which measured the mass flow $\dot{m}$ going into the wing as well as a pressure measurement p_T at the inlet of the pressure tank (as shown in Figure 2). The pipe had a diameter $D_{Exterior}$ and was measured in the wind tunnel. The pressure tank inside the wing stored the compressed the air that was pulsed through the active flow control system. Pipe 1 connected the pressure tank to the solenoids. There were 44 of these pipes that connected to each solenoid individually. The solenoid was connected to 2 nozzles each with pipe 2. Pipe 2 consisted of Pipe 2A and Pipe 2B. Pipe 2A split off into 2 pipe 2Bs at a Y-junction to connect to the 2 nozzles (as shown in Figure 3(c)). Subsequently, there were 88 pipe 2Bs and nozzle exits. Pipe 1 and 2A/B had the same diameter, D_P. At one of the nozzles, there was a pressure measurement p_K, collected from a Kulite sensor. As the air was pulsed, the measurements for pressure and mass flow would fluctuate so an average of these values was used. Temperature inside the pipe system was assumed to be equal to ambient room temperature, which was 293.15K. Kinematic viscosity was assumed to be $1.5*10^{-5}$ m^2/s in the pipe system.

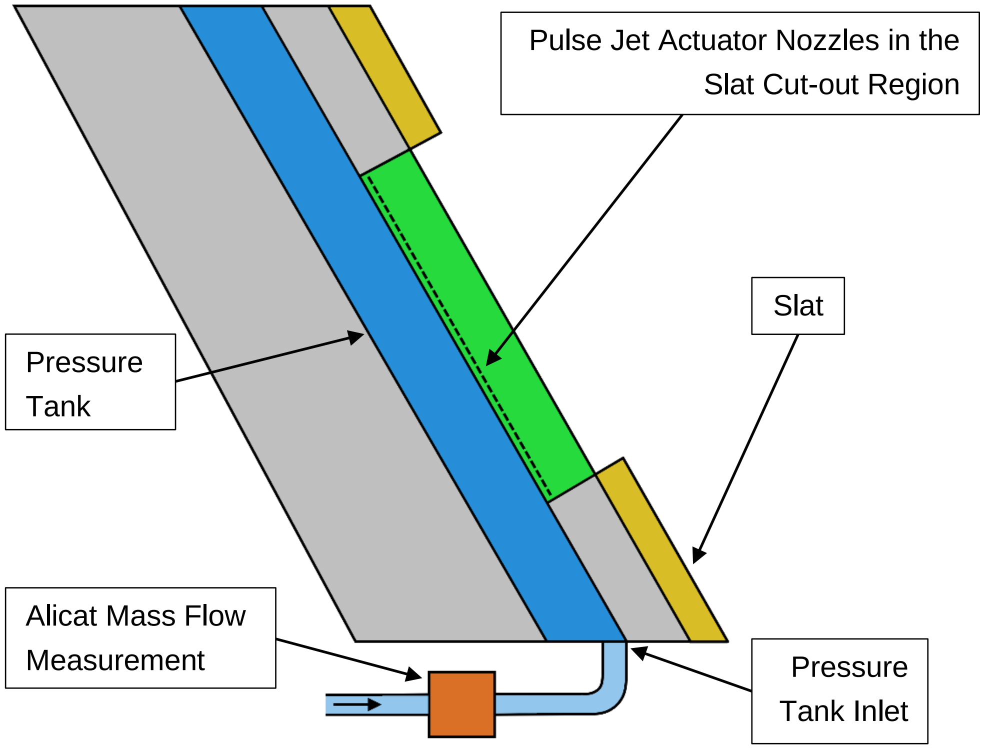

Figure 2 – WingPulse Wing Rig Setup with Pulse Jet Actuators in the Slat Cut-out Region.

A CAD model of the WingPulse flow control system was created from diagrams and dimensions obtained from documents and a CAD model (which only modelled part of the flow control system) provided by the coordinators of the project. Dimensions such as the length of pipes and diameter of pipes were found from this whereas the position of the components (such as the solenoid and pressure tank) were estimated based off of the diagrams and the known lengths of pipes. The geometry of the curved pipe (shown in Figure 3(a) as Pipe 2b) was estimated from the diagrams provided. The key geometric dimensions used in the CAD model were subsequently used in the calculations of the cost benefit analysis:

Table 2 – Geometric data used in the cost benefit analysis. Each dimension excluding the exterior pipe diameter were obtained from the creation of the CAD model.

CAD Model Measurement	Dimension value (mm)
Exterior pipe diameter	19
Pipe 1 diameter	2
Pipe 2A diameter	2

[redacted]

Pipe 2B diameter	2
Pipe 1 length	26.5
Pipe 2A length	30
Pipe 2B length	44.5

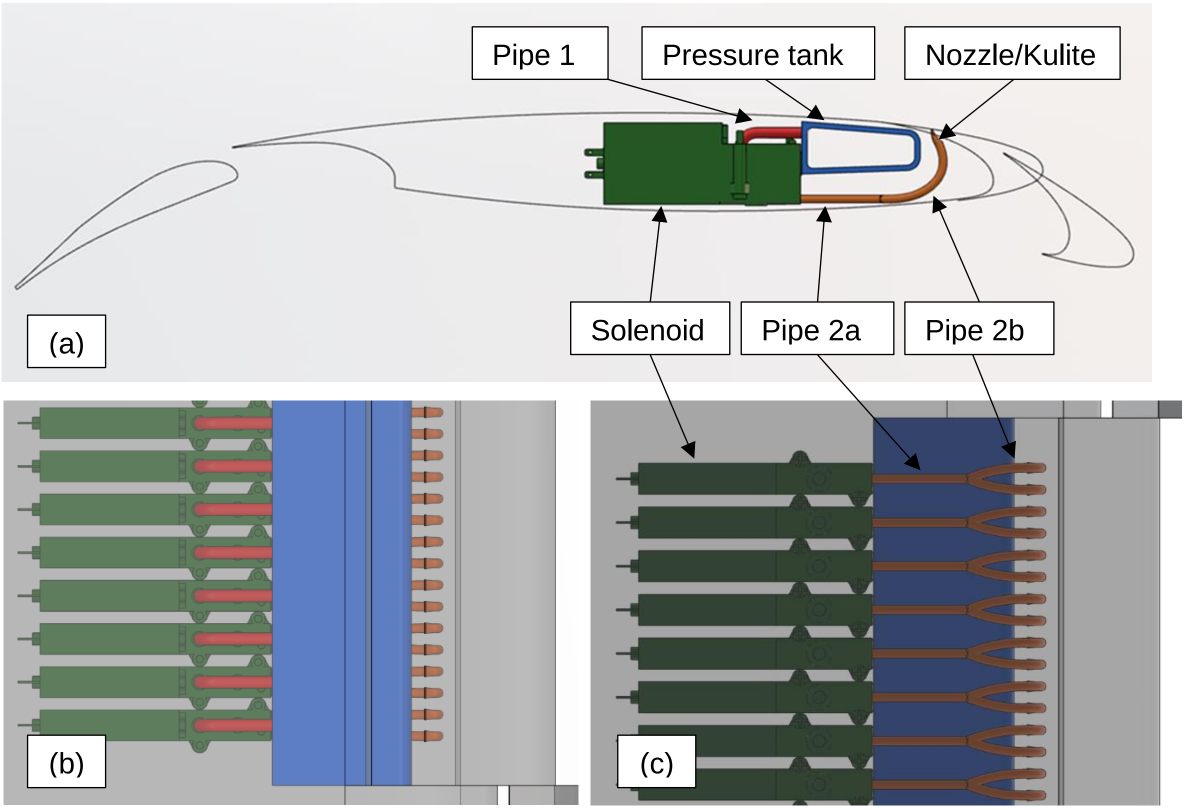

Figure 3 – (a) Sideways, (b) top, and (c) bottom view of the CAD model of the flow control system. Note that the nozzles for the pulse jet actuators are on the top of the wing.

The solenoid rig was used to test the electrical power consumption of the solenoid. This involved a single solenoid, identical to those being used in the WingPulse project, put under the same conditions it would experience during the WingPulse project. The solenoid model was a Festo MHP2-MS1H-3/2G-M5. The solenoid was connected to a power source. The power usage of the solenoid and its power source were determined by a connected device that controlled the pulse frequency and duty cycle of the solenoid. The voltage was measured by a voltmeter connected to the power source. The current was calculated as 0.1 of the voltage. The duty cycle could not be changed during the testing so it remained at 50% while the test ran at pulse frequencies of 50 and 44.1Hz. The solenoid was run for 30 seconds. The maximum and minimum root mean square voltage was recorded as the voltage would fluctuate during this time. The average root mean square voltage was calculated from this and then the peak voltage was found by multiplying the root mean square voltage by the square root of 2. The peak voltage was multiplied by 0.1 to obtain the current. The block diagram in figure 4 shows these inputs and outputs of the solenoid test:

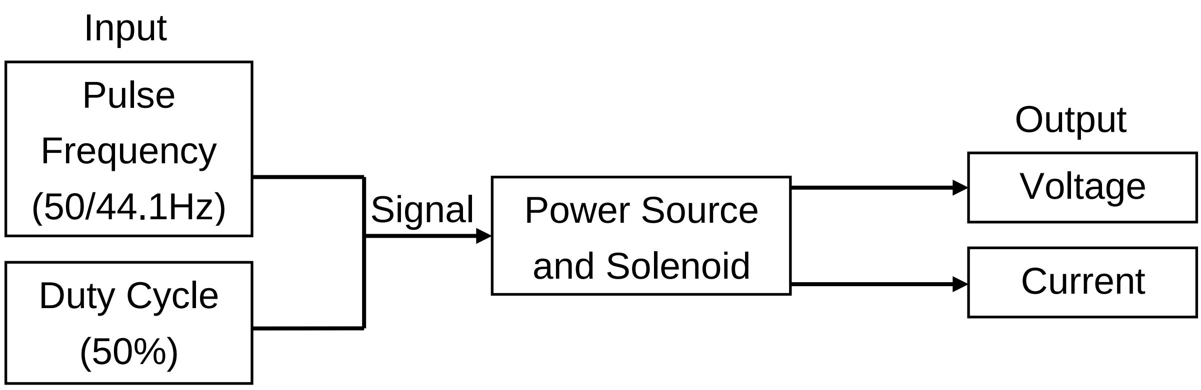

Figure 4 – Block diagram of the Solenoid experiment

3.2. Program description

The core of this project was the MATLAB script that was created to carry out the cost benefit analysis of the pulse jet actuators and, subsequently, identify the most optimal configuration for the pulse jet actuator to be in for efficiency in power consumption. Figure 4 showed the inputs and outputs of the MATLAB script:

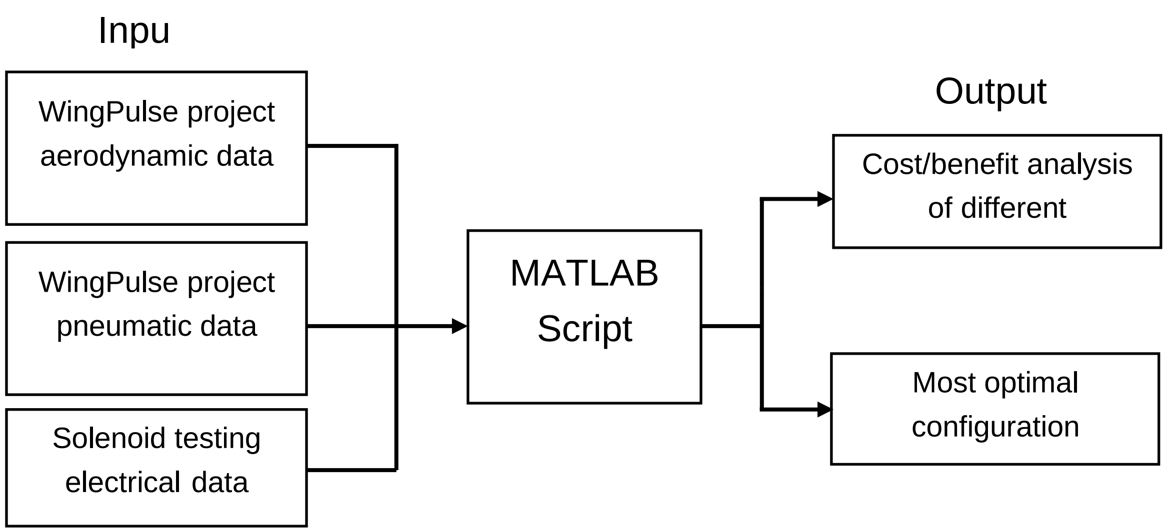

Figure 5 – Block diagram of the MATLAB Script.

The following equations were used to create the data for the pressure head losses and cost benefit analysis found in the results. The numerical values obtained for a 70% duty cycle at 14 degrees AoA from Case 1 will be shown alongside the equations for reference.

3.2.1. Pressure Head Losses

The pressure head loss of different components in the pipe system between the pressure tank and the nozzle were measured so that the efficiency of the system's design could be obtained. The pressure head loss of the solenoid and the nozzle's duct could not be accurately measured due to a lack of head loss coefficient from historic data so were neglected. The head loss coefficients obtained from historic data are presented in Table 3. Note that the value for the Y-junction was assumed to be a dividing branch flow of a T-junction:

Table 3 – Head loss coefficients of the components in the pipe system of the pulse jet actuators.

Component	Head Loss Coefficient
90° turn in pipe	0.3 [13]
180° turn in pipe	0.2 [13]
Y-junction	1.0 [13]

First, the density near the nozzle exit (1.435kg/m³) was calculated:

$$\rho_K = \frac{p_K}{R * T} \qquad \text{Equation 3}$$

Where p_K was the pressure measured near the nozzle by the Kulite (120.8kPa), R was the gas constant (287.05J/kgK), and T was the temperature inside the pipe (293.15K). The velocity was assumed to be dependent on the mass flow divided by the number of pipes that the air was moving through. At pipe 2B the air was split up into 88 pipes. With density obtained, velocity at pipe 2B (58.92m/s) was computed as:

$$V_{P2B} = \frac{\dot{m}}{\rho_K * \pi * \left(\frac{D_P}{2}\right)^2} * \frac{1}{88} \qquad \text{Equation 4}$$

Where $\dot{m}$ was the average mass flow measured at the Alicat (0.02338kg/s), D_P was the diameter of the pipe (0.002m), which remained constant between the pressure tank and near the nozzle (where the diameter changed to that at the nozzle exit). This equation assumed that there was no loss of mass flow between the Alicat (where mass flow was measured) and the rest of the pipe system. This limitation applied for every velocity calculation in the pipe system (equations 4, 10, 15, and 21). With the velocity, the Reynolds number at section 2B of the pipe was calculated:

$$Re_{2B} = \frac{V_{P2B} * D_P}{v} \qquad \text{Equation 5}$$

Where v was the kinematic viscosity in the pipe (0.00005m²/s). Reynolds number (Re = 7856) was used to calculate the friction factor of the pipe. Assuming that the pipe was smooth and 3000<Re<100000, this equation for friction factor was used [14]:

$$ff_{2B} = 0.079 * Re_{2B}^{-0.25} \qquad \text{Equation 6}$$

With friction factor (0.0084) calculated, the pressure head loss from the length of pipe 2B (0.4651kPa) was found [15]:

$$\Delta p_{2B} = ff_{2B} * \frac{L_{2B}}{D_P} * \frac{V_{P2B}^2 * \rho_K}{2} \qquad \text{Equation 7}$$

[redacted]

Where L_{2B} was the length of pipe 2B (0.0445m). The pressure head loss associated with the curve in the pipe at pipe 2B (3.736kPa) was computed, assuming that the curve resulted in a 180° turn. This was an overestimation as the curve was less than 180° [15] (shown in figure 3(a)):

$$\Delta p_{180°} = k_{180°} * \frac{V_{P2B}^2 * \rho_K}{2}$$ *Equation 8*

Where $k_{180°}$ was the head loss coefficient for a 180° turn in the pipe (0.2 [13]). The pressure head loss associated with the Y-junction in Pipe 2B (2.491kPa) was calculated [15]:

$$\Delta p_Y = k_Y * \frac{V_{P2B}^2 * \rho_K}{2}$$ *Equation 9*

Where k_Y was the head loss coefficient for a Y-junction in a pipe (1.0 [13]). As the head loss coefficient used was that of a dividing branch flow T-junction, this was an overestimation. Next the pressure head loss for pipe 2A was found, first, by obtaining the velocity in pipe 2A (117.8m/s):

$$V_{P2A} = \frac{\dot{m}}{\rho_K * \pi * \left(\frac{D_P}{2}\right)^2} * \frac{1}{44}$$ *Equation 10*

Alternatively, $V_{P2A} = 2 * V_{P2B}$. The Reynolds number in Pipe 2A (Re = 15712) was calculated:

$$Re_{2A} = \frac{V_{P2A} * D_P}{v}$$ *Equation 11*

Following this, the friction factor of the pipe (0.0071) was found, assuming that Pipe 2A was a smooth pipe:

$$ff_{2A} = 0.079 * Re_{2A}^{-0.25}$$ *Equation 12*

With the friction factor obtained, the pressure head loss was calculated for pipe 2A (1.055kPa):

$$\Delta p_{2A} = ff_{2A} * \frac{L_{2A}}{D_P} * \frac{V_{P2A}^2 * \rho_K}{2}$$ *Equation 13*

Where L_{2A} was the length of pipe 2A (0.0300m). The last pipe considered for the pressure head loss was pipe 1. Density at the pressure tank (1.132kg/m³) was calculated as:

$$\rho_T = \frac{p_T}{R * T}$$ *Equation 14*

Where p_T was the pressure measured at the pressure tank inlet (95.28kPa) and T was the temperature inside the pipe (293.15K). Pressure at pipe 1 was assumed to be the same as that at the pressure tank inlet, before the airflow passed through the solenoid which was assumed to cause a large pressure change from the pulsing of air. The velocity of the fluid in the pipe 1 (149.3m/s) was found:

$$V_{P1} = \frac{\dot{m}}{\rho_T * \pi * \left(\frac{D_P}{2}\right)^2} * \frac{1}{44}$$ *Equation 15*

The Reynolds number for pipe 1 (Re = 19910) was found with the velocity:

$$Re_1 = \frac{V_{P1} * D_P}{v}$$ *Equation 16*

The friction factor (0.0067) was found using the Reynolds number, assuming that the pipe was smooth:

$$ff_1 = 0.079 * Re_1^{-0.25}$$ *Equation 17*

Finally, the pressure head losses were obtained, starting with the calculation of the losses from the pipe's length (1.113kPa):

$$\Delta p_1 = ff_1 * \frac{L_1}{D_P} * \frac{V_{P1}^2 * \rho_T}{2}$$ *Equation 18*

Where L_1 was the length of pipe 1 (0.0265m). Then the pressure head loss associated with the 90° turn in the pipe (16.420kPa) was found:

$$\Delta p_{90°} = k_{90°} * \frac{V_{P1}^2 * \rho_T}{2}$$ *Equation 19*

Where $k_{90°}$ was the head loss coefficient of the 90° turn in the pipe (0.3 [13]). Finally, the pressure head losses were analysed and plotted to find which components caused the most losses so that the flow control system could be improved in the future to make the design more efficient and thus increase the effectiveness of the system and reduce the power consumption required. Additionally, the efficiency of the pipe layout/system (90.1%) was calculated using the total pressure head loss:

$$Pipe\ System\ Efficiency = 100 - \left(\frac{\sum \Delta p}{p_T} * 100\right)$$ *Equation 20*

Where $\sum \Delta p$ was the total pressure head loss (25.28kPa) and p_T was the pressure at the start of the system (the pressure tank inlet). The Pipe System Efficiency gave a percentage of pressure that was not lost to head losses. This percentage would need to be maximised to improve the system's efficiency.

3.2.2. Pneumatic Power

Pneumatic power was calculated from the pressurised air that entered the pressure tank in the wing. With density at the pressure tank inlet ρ_T (1.132kg/m³) obtained from Equation 14, velocity at the pressure tank inlet (72.81m/s) was computed:

$$V_T = \frac{\dot{m}}{\rho_T * \pi * \left(\frac{D_{Exterior}}{2}\right)^2} \qquad \textit{Equation 21}$$

Where $\dot{m}$ was the average mass flow rate measured at the Alicat (0.02338kg/s) and $D_{Exterior}$ was the diameter of the exterior pipe going into the tank (0.019m). Finally, the pneumatic power (62.0J/s) was obtained with the values calculated:

$$P_{Pnuematic} = \frac{1}{2} * \dot{m} * V_T^2 \qquad \textit{Equation 22}$$

3.2.3. Electrical Power

The electrical power was calculated from peak voltage and current obtained from the solenoid experiment. Note that the electrical power usage (4.2J/s) was assumed to be constant across the different duty cycles as the duty cycle could not be changed during the solenoid experiment and the electrical power consumption was insignificant compared to the pneumatic power consumption.

$$P_{Electrical} = V_P * I_P * 44 \qquad \textit{Equation 23}$$

Where V_P was the peak voltage (0.979V), I_P was current (0.0979A). The power of the single solenoid from the experiment was multiplied by the number of solenoids inside the wing in the WingPulse project (44).

3.2.4. Cost Benefit Analysis

The analysis portion of the cost benefit analysis consisted of obtaining the total power consumption of the system and comparing it to the lift and drag performance increases caused by the pulse jet actuators. This was done to observe if there was a net benefit of using pulse jet actuators and to find the most optimal configurations of duty cycle that provided the highest performance increases for the lowest power draws. This began with adding the electrical (4.2J/s) and pneumatic (62.0J/s) power consumptions to find the total power consumption (66.2J/s):

$$P_{total} = P_{Electrical} + P_{Pnuematic} \qquad \textit{Equation 24}$$

To find if there was a net gain of using the pulse jet actuators, the power consumption was directly compared with the reduction in drag coefficient by converting it into an equivalent drag consumption. First, the equivalent drag value of the power consumption (2.21N) was calculated:

$$D_{Power} = P_{total}/V_{\infty} \quad \text{Equation 25}$$

Where V_{∞} was the free stream velocity (30m/s) inside the wind tunnel. With an equivalent drag value, the equivalent drag coefficient (0.0067) was obtained:

$$C_{D\,Power} = \frac{D_{Power}}{\rho_{\infty} * V_{\infty}^2 * S} \quad \text{Equation 26}$$

Where ρ_{∞} was the free stream density (1.204kg/m³) and S was the wing area (0.306m²). Next, the change in drag coefficient between the configuration with the pulse jets turned off and the pulse jets on at the designated duty cycle was calculated to find the decrease in drag coefficient (0.038) as a result of the flow control system:

$$\Delta C_D = C_{D_{0\%}} - C_{D_{x\%}} \quad \text{Equation 27}$$

Where $C_{D_{0\%}}$ was the drag coefficient at 0% duty cycle (0.573) when the pulse jets were turned off and $C_{D_{x\%}}$ was the drag coefficient at the designated duty cycle (0.535). This was used to calculate the Figure of Merit 1 using the values obtained from the two previous Equation 26 and 27:

$$Figure\ of\ \ Merit\ 1 = \Delta C_D - C_{D_{Power}} \quad \text{Equation 28}$$

The Figure of Merit 1 value (0.031) shows whether the performance gain (specifically the drag reduction) of using pulse jet actuators had a net benefit over the power consumption. If the value was positive then the pulse jets had a net positive benefit on drag reduction. If the value was negative then the pulse jet actuators cost more in power consumption than the drag that was reduced.

Another analysis of the performance gain against the power consumption was made, using the increase in lift over power consumption as the measured metric. First, the change in lift coefficient between zero duty cycle and the designated duty cycle was calculated to find the lift increase (0.321) from using the pulse jets:

$$\Delta C_L = C_{L_{x\%}} - C_{L_{0\%}} \quad \text{Equation 29}$$

Where $C_{L_{x\%}}$ was the lift coefficient at the designated duty cycle (2.022) and $C_{L_{0\%}}$ was the lift coefficient at 0% duty cycle (1.701). With this the Figure of Merit 2 was calculated to measure the increase in lift against the power consumption:

[redacted]

$$Figure\ of\ \ Merit\ 2 = \Delta C_L/P_{total} \qquad \text{Equation 30}$$

This value (0.0049) was used in conjunction with the Figure of Merit 1 value to find the most optimal duty cycle configuration for medium (8°) and large (14°) angles of attack in terms of creating a net benefit in terms of drag reduction and their efficiency in producing lift.

4. Results and Analysis

4.1. Pressure Head Losses

Figure 6(a) shows the pressure head losses of various parts of the pulse jet actuator system against the duty cycle at 8° AoA for Case 1. The pressure head loss of the 90° turn increased by the largest amount of 3500Pa at 70% duty cycle from a minimum of 350Pa at 20% duty cycle. The second highest pressure head loss was that from the Y-junction which increased by 2100Pa from 370Pa. This was the highest pressure head loss at the lowest duty cycle (20%) but the head loss of the 90° turn became greater at 30% and above. The third highest pressure head loss was that from the frictional loss of Pipe 2 which increased by 1200Pa from 280Pa at 20% duty cycle. Pipe 1 had an increase in pressure head loss of 970Pa from 150Pa. The 180° turn in pipe had the lowest pressure head loss, with an increase of 430Pa from 74Pa.

Figure 6(b) presents the pressure head losses of the different components in the pulse jet actuator system against duty cycle at 8° AoA for Case 2. The Y-junction had the highest head losses across all duty cycles, increasing by 4000Pa from 690Pa. The 90° turn in pipe had the 2nd highest head loss from 30% duty cycle and onwards, increasing by 4200Pa from 430Pa as duty cycle increased from 20 to 70%. The pressure loss of pipe 2 was the 2nd greatest at 20% duty cycle where it was 490Pa and increased by 2200Pa. The head losses of pipe 1 and the 180° turn in pipe were the lowest, increasing by 1200Pa and 800Pa from 180Pa and 140Pa respectively.

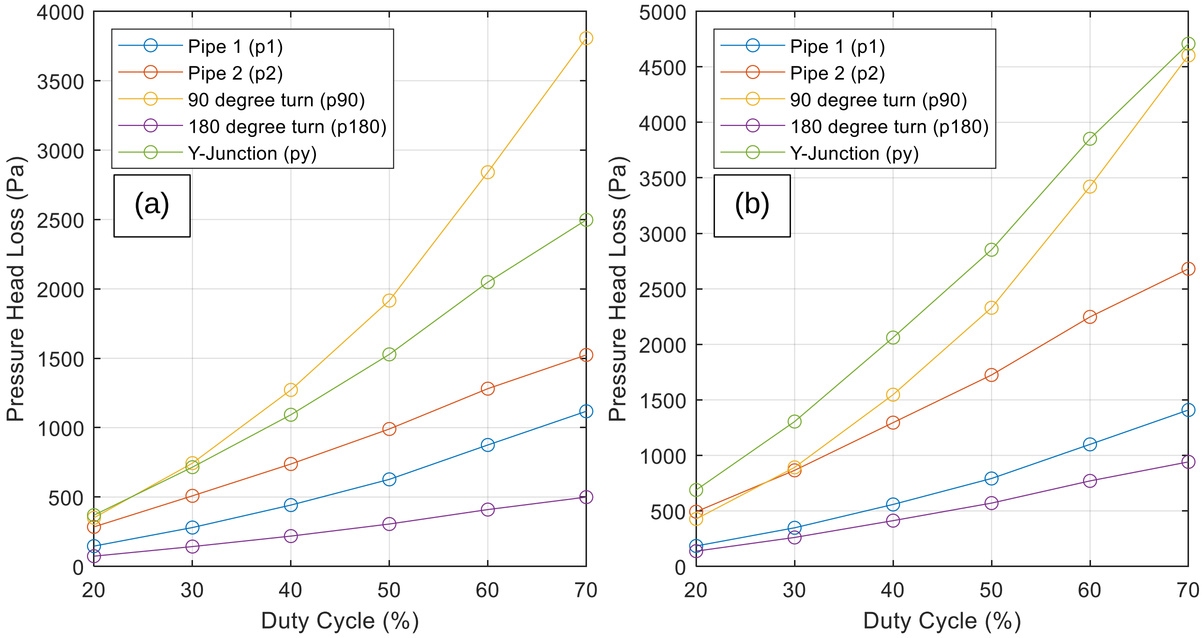

Figure 6 – Pressure Head Losses against Duty Cycle for Case 1 (a) and 2 (b).

In figure 7(a) and 7(b), the graph displays the Pipe System Efficiency (calculated using Equation 20) decreasing exponentially with increasing duty cycles for Case 1 and 2, respectively. The difference in AoA had a negligible effect on the pipe system efficiency. For both cases, system efficiency was highest at the lowest duty cycle and decreased to a minimum at the highest duty cycle. For Case 1, 20% duty cycle had an efficiency of 99.2% which decreased to 90.1% efficiency at the 70%. For Case 2, the lowest duty cycle had an efficiency of 99.2%. This decreased to 91.1% at the 70% duty cycle.

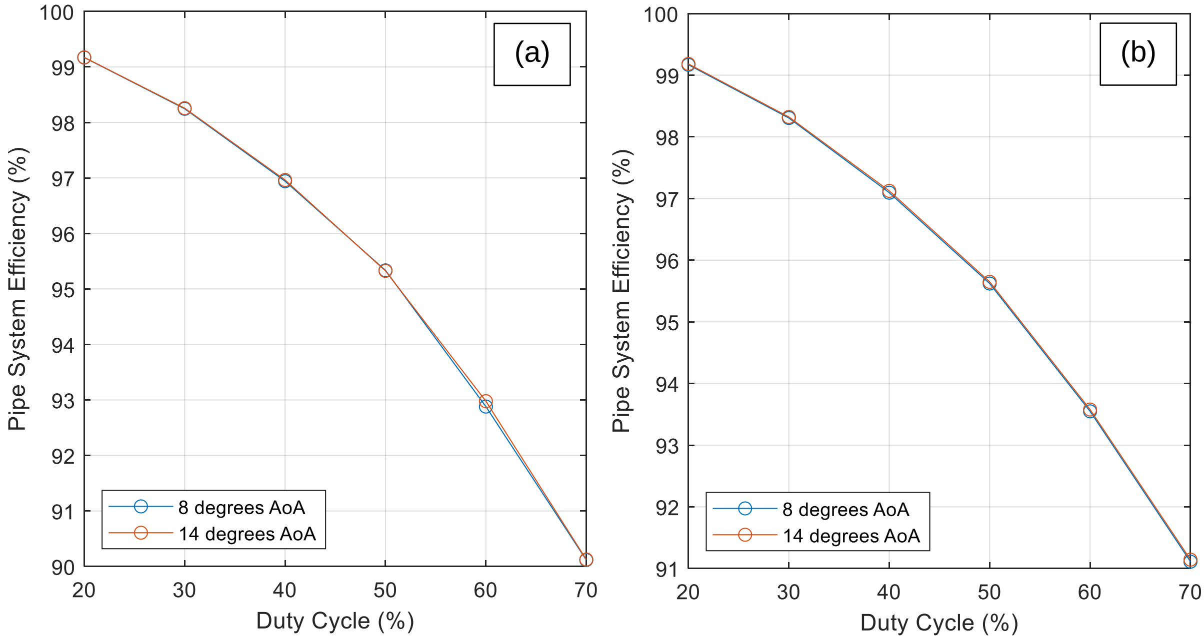

Figure 7 – Pipe System Efficiency against Duty Cycle for Case 1 (a) and 2 (b).

4.2. Aerodynamic Performance

Figure 8(a) below shows the lift coefficient across varying angles of attack for different duty cycles as well as 0% duty cycle (pulse jets are turned off) for Case 1. For 0% duty cycle, the

[redacted]

maximum lift coefficient, $C_{L_{max}}$, was 1.799 at 11° angle of attack, after which the wing entered a stall. Configurations with an active pulse jet had a peak lift coefficient at 15° angle of attack, delaying the stall by 4°. When 20% duty cycle was run at 8° AoA, the lift increased by a minimum of 0.059 (3%) while lift increased by a maximum of 0.162 (9%) at 70% duty cycle. At 14° AoA lift is increased by 0.403 (24%) at 20% duty cycle and by a minimum of 0.321 (19%) at 70% duty cycle. Maximum lift at this angle of attack occurred at a 30 to 40% when lift was increased by 0.428 to 0.429 respectively.

Figure 8(b) presents the lift coefficients of varying duty cycles across different angles of attack for Case 2. $C_{L_{max}}$ occurred at 15° AoA and varied between 2.149 and 2.089 for 40 and 70% duty cycles respectively. This caused a 0.391 to 0.331 increase in $C_{L_{max}}$ when using pulse jets, with medium duty cycles between 30 and 50% typically producing higher $C_{L_{max}}$. When 20% duty cycle was run at 8° AoA, the lift increased by a minimum of 0.073 (4%) while lift increased by a maximum of 0.188 (11%) at 70% duty cycle. At 14° AoA lift is increased by 0.421 (25%) at 20% duty cycle and by a minimum of 0.393 (23%) at 70% duty cycle. Maximum lift at this angle of attack occurred at 40% duty cycle when lift was increased by 0.442.

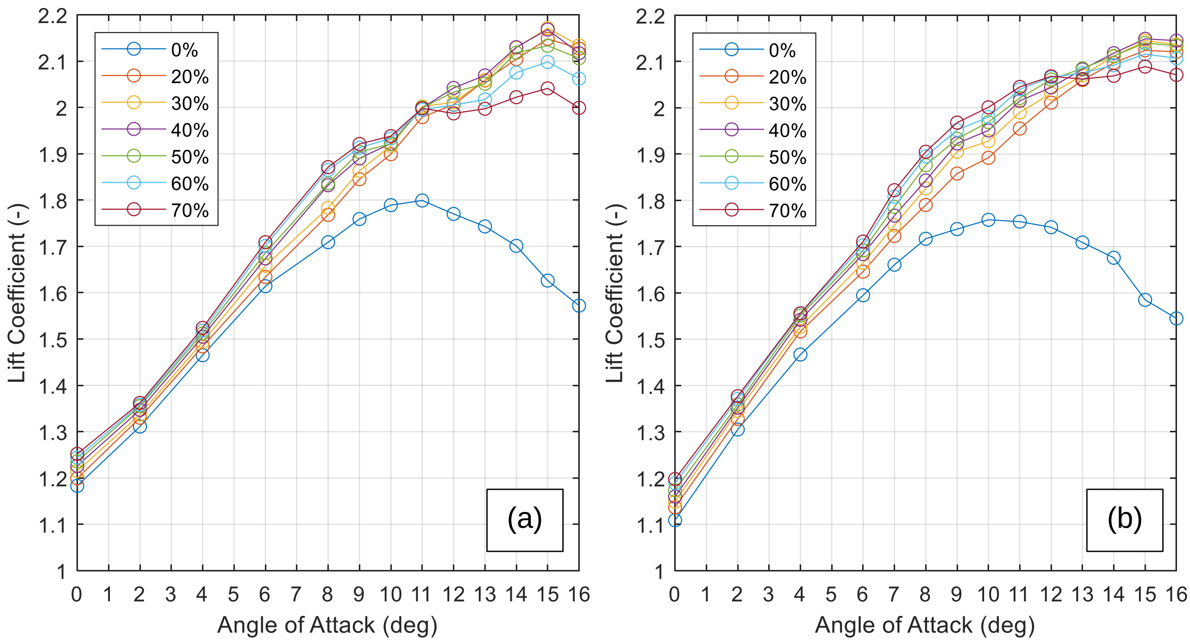

Figure 8 – Lift Coefficient against Angle of Attack for Case 1 (a) and 2 (b).

In figure 9(a), drag coefficients of different duty cycles across varying angles of attack for Case 1 are shown. When running the pulse jets at 8° AoA, drag increased by 0.002 (1%) at 20% duty cycle while an increase of 0.013 (4%) was seen at 70% duty cycle. Drag was increased by a maximum of 0.014 at 60% duty cycle and a minimum of 0.001 at 30% duty cycle. At 14° AoA, drag reduced by 0.007 (1%) at 20% duty cycle and a maximum of 0.038 (7%) at 70% duty cycle. Drag was reduced by a minimum of 0.004 at 30% duty cycle at this angle of attack.

[redacted]

Figure 9(b) shows the drag coefficients of different duty cycles across varying angles of attack for Case 2. At 8° AoA, 20% duty cycle reduced drag by 0.002 (1%) while 70% duty cycle reduced drag by a maximum of 0.020 (6%). Drag was increased by a maximum of 0.004 when the duty cycle was 30%. When running the pulse jets at 14°, the drag was reduced by a minimum of 0.018 (3%) at 20% duty cycle while drag was reduced by a maximum of 0.064 (11%) at 70% duty cycle.

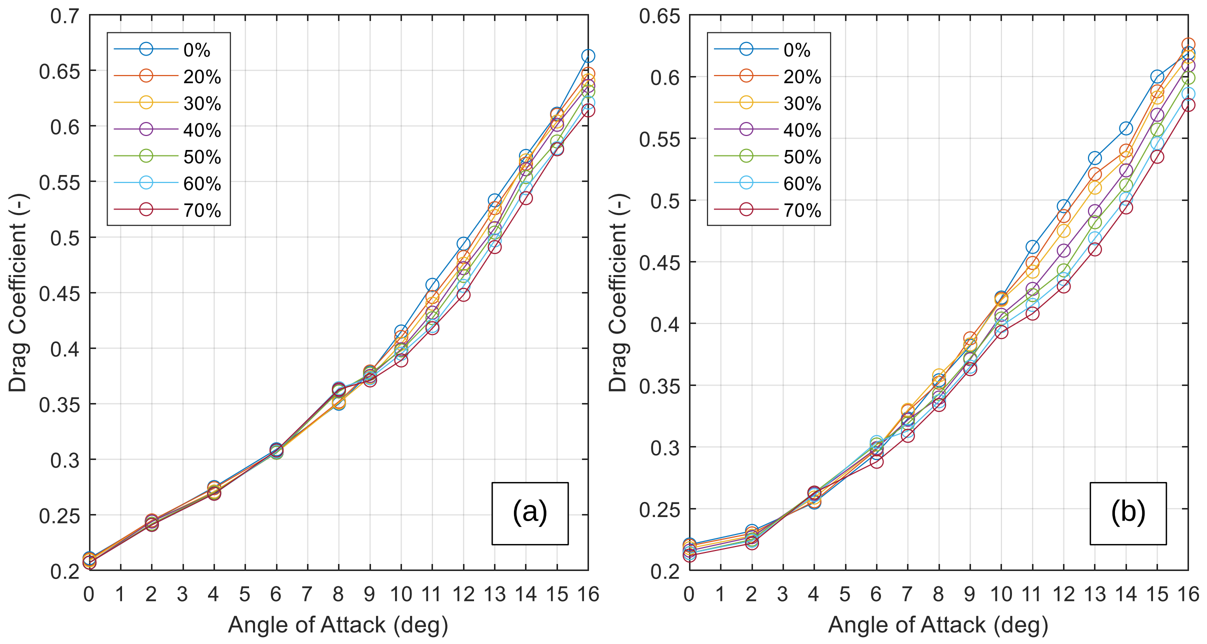

Figure 9 – Drag Coefficient against Angle of Attack for Case 1 (a) and 2 (b).

4.3. Cost Benefit Analysis

In Figure 10(a), the electrical and pneumatic power consumptions across the different duty cycles for Case 1 are shown. Electrical power consumption was constant at 4.2 J/s while pneumatic power increased exponentially from 1.4 J/s at 20% duty cycle to 62 J/s at 70% duty cycle. Pneumatic power surpassed electrical power consumption at 30% duty cycle.

Figure 10(b) presents the electrical and pneumatic power consumptions of the pulse jet system across the different duty cycles for Case 2. Electrical power consumption was constant at 3.9 J/s while pneumatic power increased exponentially from 1.5 J/s at 20% to 64 J/s at 70% duty cycle. Pneumatic power consumption exceeded electrical power consumption at 30% duty cycle.

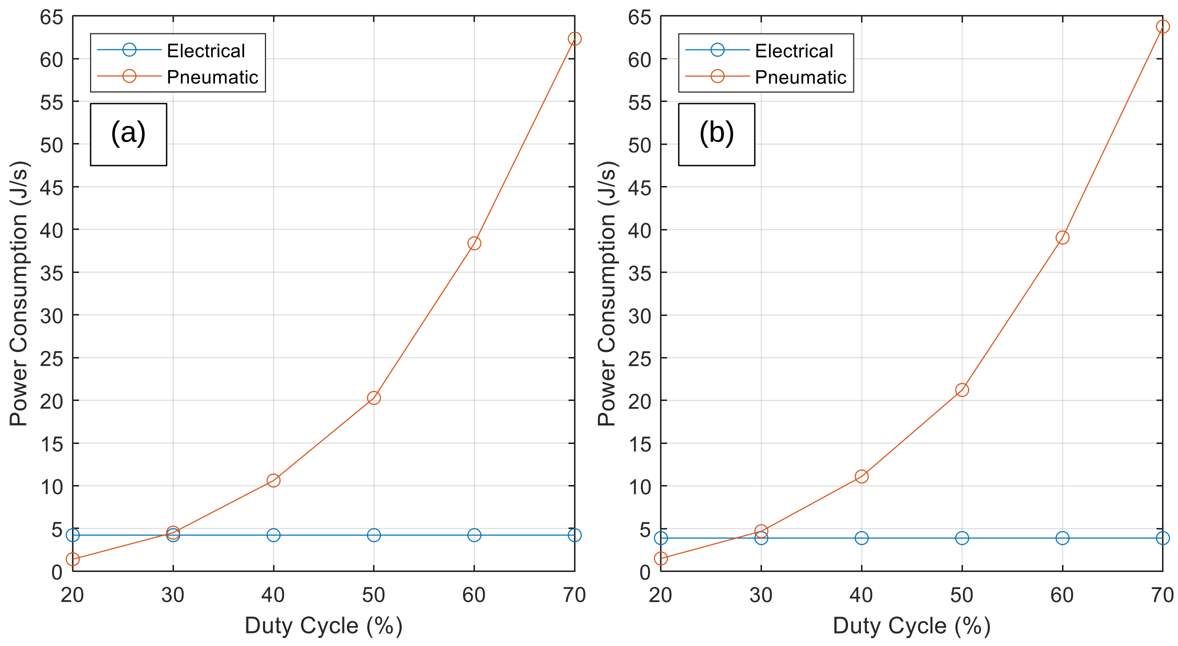

Figure 10 – Power Consumption against Duty Cycle for Case 1 (a) and 2 (b).

Figure 11(a) shows two graphs from two different angles of attack (8 and 14°) with the Delta C_d - Power C_d (Figure of Merit 1 calculated in Equation 28) against duty cycle for Case 1. At 8° AoA, the Figure of Merit 1 increased to its peak of -0.001 at 30% from -0.003 at 20% then decreased with increasing duty cycle to its minimum value of -0.020 at 70% duty cycle. Oppositely, at 14° AoA, the Figure of Merit 1 initially decreased to its minimum value of 0.003 at 30% duty cycle before increasing steadily as duty cycle increased, meaning that the value was always positive at this AoA. The graph peaks at its highest duty cycle of 70% to a value of 0.031.

In Figure 11(b), two graphs from the 8 and 14° angles of attack with the Delta C_d - Power C_d against duty cycle for Case 2 are displayed. When running the pulse jets at 8° AoA, the Figure of Merit 1 decreased to its minimum of -0.005 at 30% duty cycle from 0.001 at 20% and then increased with duty cycle to a maximum of 0.013 at 70% duty cycle. At 14° AoA, the Figure of Merit 1 started at its minimum of 0.012 at 20% duty cycle before increasing with duty cycle to a maximum of 0.067 at 70% duty cycle, thus remaining positive in value across all duty cycles.

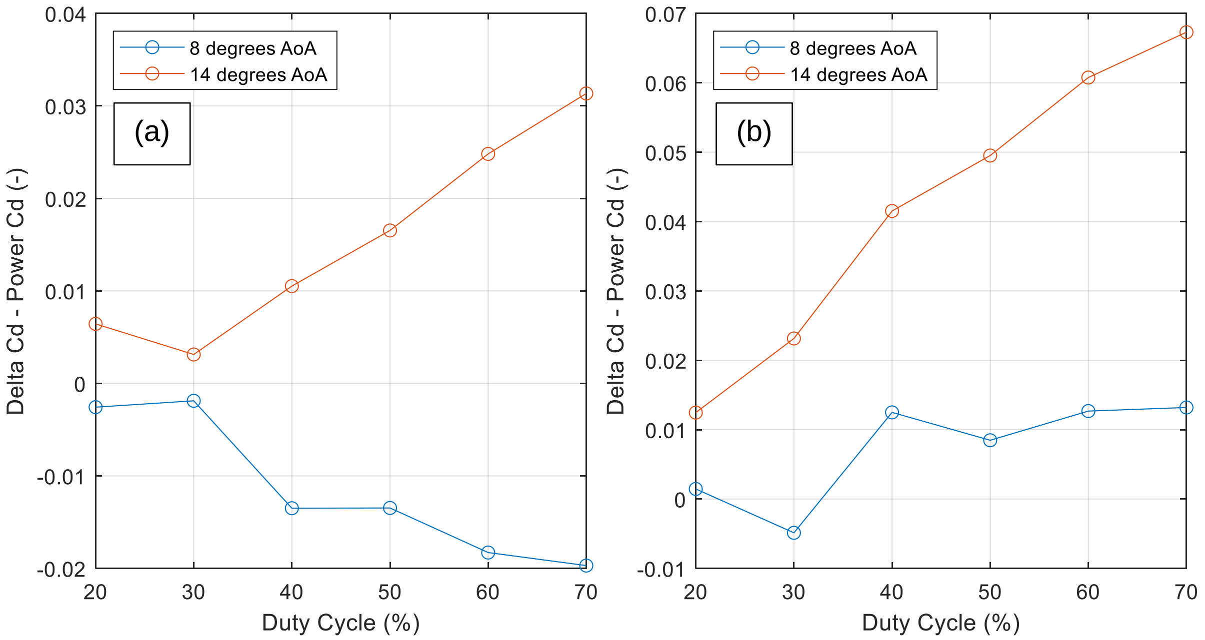

Figure 11 – Delta C_d - Power C_d against Duty Cycle for Case 1 (a) and 2 (b).

Figure 12(a) displays Delta C_L / Power (Figure of Merit 2 calculated in Equation 30) against duty cycle for two different angles of attack (8 and 14°) for Case 1. In the graph for 8° AoA, the Figure of Merit 2 steadily decreased from 0.010 at 20% duty cycle to 0.008 between 30 and 40% duty cycle before decreasing at slightly increased rate to a minimum of 0.002 at 70% duty cycle. At 14° AoA, the Figure of Merit 2 initially peaked at 0.072 at 20% duty cycle before decreasing exponentially to a minimum of 0.005 at 70% duty cycle.

In Figure 12(b), the Delta C_L / Power at 8 and 14° AoA is plotted against duty cycle for Case 2. When running the pulse jets at 8° AoA, the Figure of Merit 2 was at its maximum of 0.014 at 20% duty cycle before decreasing to its minimum of 0.003 at the highest duty cycle. At 14° AoA, the Figure of Merit 2 begun at a maximum of 0.066 at 20% duty cycle before decreasing exponentially to a minimum of 0.005 at 70% duty cycle.

[redacted]

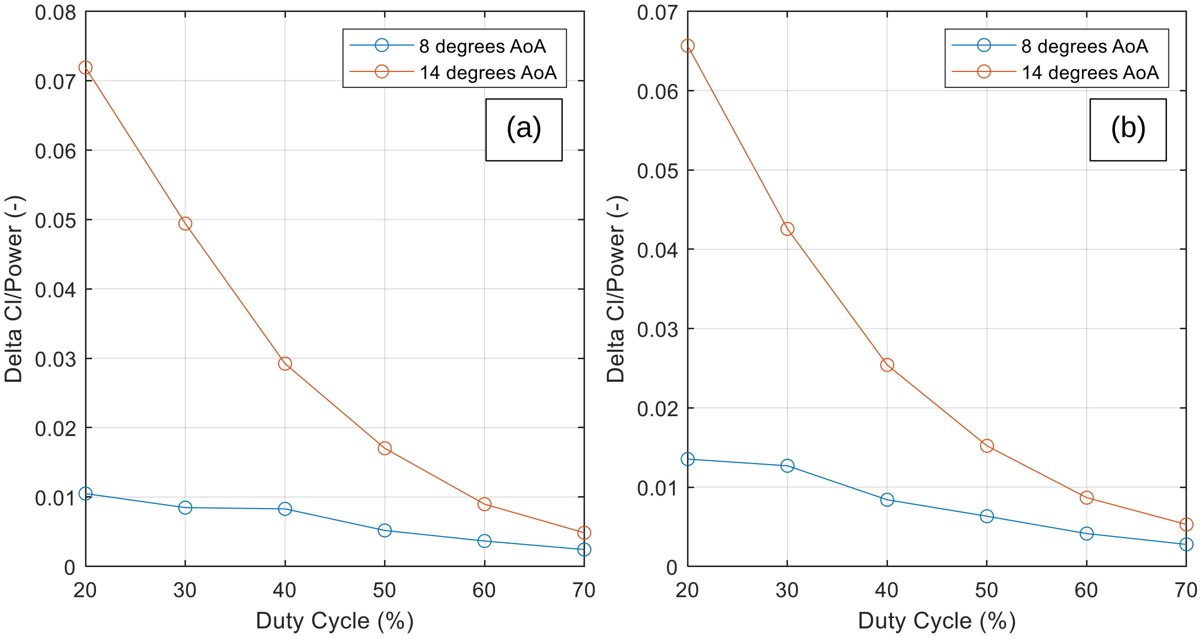

Figure 12 – Delta C_L / Power against Duty Cycle for Case 1 (a) and 2 (b).

5. Discussion

5.1. Pressure Head Losses

The results from the pressure head loss graphs (Figure 6) showed that the losses for the 90° turn in pipe and Y-junction were the highest of the component head losses in the pipe. The Y-junction had a high head loss because it occurred from geometry of the pipe (minor loss) rather than the pipe's roughness (major loss). This was smaller than that of the geometrical changes as the pipe was assumed to have a small friction factor while also having a relatively small length over diameter ratio for each section of pipe. As a result the friction factor multiplied by the length over diameter ratio was smaller than the head loss coefficients of Y-junction – which had a significantly higher loss coefficient than the other components because it was assumed to be a T-junction. This was an overestimation as a T-junction would produce more pressure head loss than a Y-junction. For the 90° turn in pipe, the velocity of the airflow at this point in the pipe was double that of the velocity at the other two geometrical changes that were accounted for. The velocity was squared in the pressure head loss calculations (Equation 7,8,9,13,18, and 19) so a greater velocity value caused an exponential change in head loss. Velocity at this turn in pipe was double that at the other geometrical pipe losses, where the pipe flow was split into 2 and thus the mass flow and velocity was halved. This was a limitation of the experiment as the velocity could not be measured elsewhere in the pipe system. The density of the fluid in the pipe was only calculated at 2 points – at the location of the Kulite sensor near the nozzle and at the inlet to the pressure tank. This added further limitations to the calculations of the velocity in the pipe, particularly at higher duty cycles where velocities exceeded 100m/s in some parts of the pipe system and thus the density may not have been constant in the system due to compressibility. The higher velocity squared resulted in much higher increases in the pressure head loss of the 90° turn in pipe as duty cycle increased. At low duty cycles, the velocity was low and the head loss of this component was relatively low as the loss coefficient was small. Velocity in the pipe was calculated using the mass flow, measured at the Alicat, and dividing it by how many pipes the flow was going through and the area of the pipe (Equation 4,15, and 21). The Y-junction had a higher loss than the turn in pipe in Case 2 for all duty cycles as pressure was higher. This caused density to be greater and have a bigger impact than velocity on the head loss. At the highest duty cycle, the 90° turn in pipe was beginning to merge with the Y-junction losses as the velocity was increasing exponentially. In Case 1, where pressure and thus density was lower, the Y-junction only had the highest pressure head loss at the lowest duty cycle. The frictional loss of pipe 2 caused the third greatest pressure head losses for both cases as the length of pipe was more than double the length of pipe 1. The 180° turn in pipe had the lowest pressure head loss as the head loss coefficient was smaller than the 90° turn head loss coefficient as it had a smoother bend compared to the sharp elbow of the 90° turn in pipe. The velocity was also half of that of that at the 90° turn in pipe. Mass flow/velocity increased with duty cycle, resulting in an exponential increase in pressure head loss of the all the components with increasing duty cycle. The pressure head losses were greater in Case 2 as the higher pressure caused a higher mass flow/velocity and density.

From the data gathered in the pipe system efficiency graphs (Figure 7) it was shown that the overall efficiency of the pipe system decreased exponentially as duty cycle increased. This was because the overall pressure head loss increased as the velocity increased with duty cycle. As duty cycle increased, the average mass flow through the pipe system increased and therefore the velocity increased. As the head loss is not dependent on any variables outside the wing, the AoA had a negligible effect on the pipe system's efficiency. As a result, it might be beneficial to minimise the duty cycle to maximise the efficiency of the system so power consumption waste is minimised. When running the flow control system at the higher pressure setting in Case 2, the efficiency of the pipe system was the same as in Case 1 until it reached 70% duty cycle where the system was 1% more efficient. This suggests that running the pipe system at a higher pressure while running the high duty cycles would make the pipe system more efficient however, as seen from Figure 6, the overall pressure head loss would also be higher and the increase in system efficiency would be relatively small. The efficiency increased in Case 2 because the increase in pressure over Case 1 was more than the increased head loss. Additionally, the system efficiency could be improved by reducing pressure head loss of components of the pipe system. As the 90° turn in pipe had the highest head loss, removing or reducing this component could cause a significant boost to the efficiency of the pipe system. This could be done by redesigning the system so that the turn in pipe is not necessary or reducing the fluid flow in the system as velocity increases pressure head loss by a factor of x^2 (as seen in Equation 18). However, reducing fluid flow may have a detrimental impact on the performance of the pulse jets as a result of a reduced velocity ratio. From historic data gathered in the literature review, it is said that a velocity ratio below 1.6 would decrease the effectiveness of the pulse jets, with the flow becoming non uniform from being disturbed from the cross flow. As the velocity ratio used in this study was 4.2 and 5 for Case 1 and 2 respectively, the velocity of the pulse jets could be reduced significantly before a large impact on performance would be seen according to this historic data. Alternatively, the 90° bend in pipe could be made smoother so it was not an elbow curve and was a more gradual, long radius bend. This would reduce the head loss coefficient slightly and decrease the pressure head loss at the bend. The pipe system efficiency could be improved further by altering the other components in the system, with the Y-junction head losses then the frictional losses of pipe 2 being of highest importance to reduce after the 90° turn in pipe head loss had been minimised.

5.2. Aerodynamic Performance

The results shown in the lift coefficient against AoA graph for the two cases (Figure 8) indicate that the pulse jet actuators, of duty cycle between 20-70%, cause a delay in the stall of the wing by 4° AoA – from 11° to 15°. This correlates with historic data that found that pulse jets do help cause a delay in stall. For Case 1 at 14°, lift was increased by 24% at the lowest duty cycle while this reduced to 3% at 8°. Lift increased by 25% at 14° and 4% at 8° AoA when the pulse jets were run at 20% for Case 2. For the highest duty cycle, lift was increased by 19% at 14° but this reduced to a 9% increase at 8° AoA in Case 1. For Case 2, lift increased by 23% at 14° and 11% at 8° when the duty cycle was 70%. This implied that running a lower duty cycle at high angles of attack, such as at 14°, would be the most optimal configuration in producing the most lift while minimising power consumption at the same time. Lower duty cycles caused a lower mass flow requirement as less air was being pushed out of the pulse jet actuators per second, reducing the pneumatic power consumption. This was promising for this study which aimed to find the most optimal configuration of pulse jets with performance against power consumption being considered. From historic data, this higher performance at lower duty cycles may have occurred because as duty cycle increased, vortex rings produced by the pulsed flow would move closer together and interact with each other, causing the destruction of these vortex rings, resulting in turbulent flow. However, running the pulse jets at lower angles of attack might not have been beneficial given the small impact on additional lift it produced in this flight regime, particularly in the case of the slat cut-out region where the primary aim was to delay stall and increase lift and high AoA as a slat would.

From the data gathered in the drag coefficient against AoA graph for both cases (Figure 9), it was shown that the pulse jet actuators reduced drag at angles of attack greater than 9° for both cases while having a negligible or even a negative impact on drag at lower AoA – especially at 8° in Case 1 where drag coefficient increased by over 0.01 for duty cycles above 30%. This was unlike the lift coefficient graphs (Figure 8) where running the pulse jet actuators always increased lift, regardless of the AoA, meaning that the pulse jets would always have a positive impact on the lift performance, even if it was relatively small at lower angles of attack. At 8°, the lowest duty cycle of 20% had the smallest increase in drag of 1% but also reduced drag at 14° by 1% in Case 1. The performance of the lowest duty cycle was

[redacted]

slightly improved in Case 2 when drag was reduced by 1% at 8° and 6% at 14°. At 14° AoA in Case 1, lift increased by a maximum of 25% (30-40% duty cycle) when pulse jet actuators were used whereas drag reduced by a maximum of 7% (70% duty cycle). At the same AoA for Case 2, lift similarly increased by a maximum of 26% (40% duty cycle) while drag was reduced by a maximum of 11% (70% duty cycle). As the greatest drag reduction was seen to be produced by the biggest duty cycle and higher pressure case, the efficiency of using pulse jets to reduce drag was less than that of using pulse jets to increase lift where lower duty cycles increased lift more at high AoA. As a result, measuring the effectiveness of the pulse jets off of the drag reduction would not be the best representation of the performance increases that the pulse jets caused, being considered a side effect to the main performance increase in lift at high AoA. This was seen in previous experiments done on pulse and steady jet actuators where lift increase was the primary performance variable measured.

5.3. Cost Benefit Analysis

From the data presented in the power consumption against duty cycle graphs for both cases (Figure 10), the pneumatic power was seen to increase exponentially with duty cycle. This was because the average mass flow increased with duty cycle and velocity, which was a function of mass flow, was squared in the pneumatic power equation (as shown in Equation 22). As a result of this, the performance of the pulse jets at higher duty cycles would have had to have been exponentially higher than at the lowest duty cycle to be more efficient in the cost benefit analysis.

As shown from the results in the Figure of Merit 1 against Duty Cycle graph for Case 1 and 2 (Figure 11), running the pulse jet actuators at higher angles of attack (of 14° in the graph) would result in a net benefit of drag reduction C_D minus the power consumption equivalent C_D (Equation 28). This means that the pulse jet actuators reduced drag more than they 'created' it through power consumptions. As discussed in the aerodynamic performance, at high AoA, the greatest drag reduction was produced by the largest duty cycles. This was reflected here when the Figure of Merit 1 increased with duty cycles - suggesting that the increase in power consumption was less than the increased drag reduction when duty cycle increased above 30%. The greatest net benefit from drag reduction occurred at the highest duty cycle of 70% (between 0.031-0.067 for Case 1 and 2, respectively), suggesting that pulse jet actuators were most effective in drag reduction when more air was pulsed (at high angles of attack) and the increased equivalent drag from the higher power consumption was less than that of the increased drag reduction of the highest duty cycles. In Case 2, the Figure of Merit 1 was greater than in Case 1, with the value of the Figure of Merit 1 more than doubling in value over its Case 1 equivalent as duty cycle increased to 70%. This suggested that running a higher pressure greatly increased the efficiency of drag reduction at higher angles of attack.

In Case 1 at the lower AoA of 8°, there was a negative net benefit of using the pulse jets for drag reduction for all duty cycles. At 8° AoA, duty cycles produced slightly more drag than when the pulse jets were switched off. This meant that having pulse jet actuators turned on

[redacted]

at this angle was negatively impacting the drag performance while needlessly consuming power. The Figure of Merit 1 reached its peak at 30% duty cycle (-0.001). As explained in the aerodynamic performance, this was where drag was increased by the smallest amount, with 20% duty cycle producing fractionally more drag. At duty cycles greater than 30%, the drag increased by a relatively large amount and thus the graph fell sharply while power consumption also increased, further decreasing the value of the Figure of Merit 1 at these higher duty cycles. Case 2 presents a different graph for this AoA, with the Figure of Merit 1 increasing as duty cycle increased. Apart from 30%, every duty cycle caused a decrease in drag, with duty cycles above 40% producing a relatively large drag reduction (reaching a maximum of 0.013 at 70% duty cycle). The Figure of Merit 1 increased with duty cycle at a very low rate compared to when the pulse jets were run at 14° AoA. This suggested that despite highest duty cycles having the greatest impact on drag reduction, the greater power consumption negated the increase in performance as this performance increase was relatively small. When comparing the two cases at 8°, it was implied that running the pulse jets at a higher pressure would increase the efficiency of drag reduction at medium angles of attack, particularly for medium to higher duty cycles.

The data shown from Figure 12 presents that at 14° AoA, the lift increase was most power efficient at the lowest duty cycles of 20% (reaching 0.072 and 0.066 for Case 1 and 2, respectively), with a sharp decline in Figure of Merit 2 for both cases after this point. This was because the lower duty cycles between 20 and 50% had the highest increases in lift at 14°. At duty cycles higher than 20%, the lift increases were outweighed by the power consumption required by increased duty cycles – at 14° AoA, 30, 40 and 50% duty cycles had a higher lift increase over the 20% duty cycle, but the increased power consumption requirements resulted in a Figure of Merit 2 decreasing past 20% for both cases.

Figures 12 demonstrated that at the lower angle of attack of 8°, the maximum lift efficiency was drastically lower than the maximum efficiency at 14° AoA, being as much as 85% lower in lift efficiency. For both cases, the Figure of Merit 2 begun at a maximum at 20% duty cycle (0.010 and 0.014 for Case 1 and 2, respectively) before decreasing to its minimum at 70% duty cycle. This occurred at a much slower rate when compared to the decreases at 14°, which implied that duty cycle had a smaller effect on lift efficiency at this AoA. The huge decrease in lift efficiency at 8° suggested that running pulse jets at lower angles of attack was not advantageous given the relatively small performance gains it provided while requiring the same power consumption levels as those at higher angles of attack where performance gains were significantly greater. As a result, it might have been beneficial for power consumption savings to run the pulse jets exclusively at high angles of attack, particularly at angles of attack above the stall angle of the wing (in this case 11°) and at the lowest duty cycle of 20% where lift was produced most efficiently according to the Figure of Merit 2 in Figure 12.

6. CONCLUSIONS

The aim of this project was to conduct a cost benefit analysis to study the energy consumption of using pulse jet actuators for active flow control in the slat cut-out region to facilitate Ultra High Bypass Ratio engines. The cost benefit analysis was run using data collected from two physical rigs – the WingPulse project and the solenoid experiment – as well as the CAD model of the flow control system that was created. The analysis calculated pressure head losses of the pipe system inside the pulse jet actuators, along with lift increases and drag reduction caused by the pulse jets compared against the total power consumption generated by running the flow control system. Key findings from this analysis include:

- The 90° turn in pipe and Y-junction generated the most head loss of all components in the pipe systems for both cases.
- Pipe efficiency was highest at the lowest duty cycle (99.2%) while it was lowest at the highest duty cycle (90.1-91.1% for Case 1 and 2, respectively).
- 70% duty cycle at a high AoA (14°) had the greatest net benefit in reducing drag between 0.031-0.067 Delta C_d - Power C_d for Case 1 and 2, respectively.
- 20% duty cycle at a high AoA was the ideal configuration for the most efficient lift increases between 0.072-0.066 Delta C_L / Power for Case 1 and 2, respectively.
- The pulse jet actuators had the lowest net benefit in drag reduction and lift efficiency at a lower angle of attack (8°), with net benefit in drag as low as -0.020 and lift efficiency at a minimum of 0.002 at 70% duty cycle.

7. FUTURE WORK

The next step for this work would be to build on this study by improving the methodology while continuing tests on the same wing model. This could be done with a CFD model of the pipe system which would accurately simulate the flow to obtain more accurate head losses. This would also allow for a better understanding of how the pulsed air behaves inside the pipes to find ways to increase the efficiency of the system and thus reduce the power consumption requirements. In addition to this, a further study could also include the implementation of weight and space into the cost benefit analysis, as these variables are very important for implementing the flow control system onto a real wing with weight and space restrictions. This future study could evaluate more frequency, velocity ratio and pressure variables with the existing duty cycle configurations at additional angles of attack to gather more data on which variable setup is most power efficient in the context of existing literature of different configurations for the pulse jet actuators. Following this, the pipe system could be improved to increase the efficiency of the design by changing the components that caused the greatest head losses.

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